The Cross-Factor Correlation Mismatch in Paper 02's Simulation Driver: `c.cfct` Set to Half of `c.cf1t` Where the Refined Process Specifies One Constant

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Purpose. This note describes one specific defect found on 2026-10-21 in `analysis/scripts/carryover-sensitivity/simul` and one other driver script (`analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.R`), the data-generating drivers behind paper 02 ("A factorial comparison of nine carryover representations for aggregated N-of-1 trials under covariance moderation"). The refined process specifies the correlation between different response components, at any pair of timepoints including the same visit, as a single separable form, $c_1\,\rho^{|w_t - w_s|}$, with $c_1 = 0.2$ (paper 01, Table `tab:spec`; `docs/series/ch10-refined-process.md`). The driver's two model parameters that jointly implement this formula, one for same-timepoint pairs and one for different-timepoint pairs, were set to two different values, so every cross-factor correlation beyond the same visit was exactly half the documented value, for the entire production history of paper 02 until this session's fix. This note states the intended model, quotes the code as it stood, derives the exact size of the discrepancy, works a numeric example with verified values, states what is and is not affected, gives the fix verbatim, and records what remains open. It is written for a reader with statistical training (correlation matrices, variance components) but no prior exposure to this project's code, in the style of `docs/53` and `docs/55`, which document analogous defects in Hendrickson et al.'s published code; this defect is instead in this project's own code, not in that external reference. Section 3 uses the Kronecker-product (separability) framing developed at length in `docs/63-why-correlation-fades-with-time-a-clinicians-guide.md`; a reader unfamiliar with that framing may want to read Section 3 of that note first, though this note restates what is needed to follow the bug itself.

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Notation index and glossary

Notation follows the project's canonical table (`analysis/report/NOTATION.md`) where applicable. Symbols marked *local* are defined here for this note only.

Table 1: Notation index for this note

Symbol	Meaning	Status
ρ	within-factor AR(1) autocorrelation per week (code `c.tv`, `c.pb`, `c.br`, here always 0.7 in the affected cells)	canonical
w_t, w_s	calendar week of visit t, s	canonical
c_1	the refined process's single cross-factor correlation constant (paper 01 Table `tab:spec`)	canonical
`c.cf1t`	code parameter: cross-factor correlation at the same timepoint (lag 0)	local (code)
`c.cfct`	code parameter: cross-factor correlation scale at different timepoints, multiplying ρ^{lag}	local (code)
TV, PB, BR	the three response components: natural course, placebo belief, biological drug response	canonical
c_{bm}	on-drug biomarker-BR correlation (code `c.bm`); unaffected by this defect	canonical
"the affected driver"	`simulation-core.R`'s `simulate_cell()` and `simulate_cell_s7()`, and `26-co-dbc-fix-paired.R`'s `run_cell()`	local
"the shared function"	`build_correlation_matrix()` in `implementations/tidyverse/R/functions.R`, mirrored by `buildSigma()` in the package's `R/generateData.R`	local

Glossary.

- **Cross-factor correlation.** The correlation between two *different* response components (e.g. TV and PB) at a given pair of visits, as distinct from the within-factor AR(1) correlation between the same component at two visits.
- **The refined process.** The pmsimstats-ng data-generating process recommended by paper 01 as the reference DGP for simulation studies of this kind (`NOTATION.md`; paper 01 Table `tab:spec`).
- **The affected driver.** The caller-side code that builds the `model_param` list passed into the shared function, not the shared function itself (Section 4 explains why this distinction matters here).

Evidence labels. Each claim carries one of four labels: *quoted* (text reproduced from a source), *inspected* (read from source code), *derived* (follows algebraically) or *verified* (computed by running code, with the exact commands given in Section 9).

1. Summary for the busy reader

What the code did. `simulation-core.R`'s two model-parameter constructors (`simulate_cell()`, line 467 as now fixed, and `simulate_cell_s7()`, line 878 as now fixed) and `26-co-dbc-fix-paired.R`'s `run_cell()` (line 130) set `c.cf1t = 0.2` but `c.cfct = 0.1` (inspected, prior to this session's fix). The shared function that consumes these, `build_correlation_matrix()`, uses `c.cf1t` directly for same- timepoint cross-factor pairs and `c.cfct * rho^lag` for every other pair (inspected, `implementations/tidyverse/R/functions.R` lines 236-258).

What it should have done. The refined process specifies a single constant $c_1 = 0.2$ scaling $\rho^{|w_t - w_s|}$ at *every* lag, including lag 0 where $\rho^0 = 1$ makes the formula collapse to exactly c_1 (quoted, paper 01 Table `tab:spec`: "Cross-component correlation | separable, $c_1\rho^{|w_t-w_s|}$ […] | $c_1 = 0.2$";

`docs/series/ch10-refined-process.md` line 115 states the same form for the refined process against the published process's constant $c_{\times} = 0.1$ at every lag). The code's two-parameter structure is general enough to represent this single-constant form as a special case — it simply requires `c.cf1t` and `c.cfct` to be set to the same value. They were not.

How large. Because the intended and the actual value both scale by the same ρ^{lag} factor, the ratio of actual to intended is exactly `c.cfct` / c_1 = 0.1 / 0.2 = **0.5** at every lag greater than zero, independent of ρ and of the specific lag (derived). At the same timepoint (lag 0), the correlation was correct throughout, since that case uses `c.cf1t` directly, which was set to 0.2 as documented. Concretely, at a 4-week gap with $\rho = 0.7$: the intended TV-PB correlation is $0.2 \times 0.7^4 = 0.04802$; the code produced $0.1 \times 0.7^4 = 0.02401$ (verified, Section 6).

What this is not. This is not a defect in the shared function `build_correlation_matrix()` (or the package's `buildSigma()`, which implements the identical two-parameter structure). Both functions correctly implement a general separable cross-factor form; the defect is entirely in what the caller passed for `c.cfct`. No change to the shared function was needed or made.

What it does and does not touch. The within-factor AR(1) correlations (`c.tv`, `c.pb`, `c.br`) and the biomarker-BR differential coupling `c.bm` are generated by separate code blocks and are unaffected (inspected). What is affected is the background correlation among the three nuisance/signal components (TV, PB, BR) with each other across different visits — every cross-timepoint, cross-factor entry of the covariance matrix, in every design with more than one visit, which is every design this manuscript uses (Hybrid, OL+BDC, CO all have multiple visits). At the one reference cell checked (Hybrid, $c_{bm} = 0.45$, $t_{1/2} = 1.0$), the resulting covariance matrix remained positive definite under both the old and the corrected parameterization, though the minimum eigenvalue was smaller under the old, weaker-correlation parameterization (0.1025) than under the corrected one (0.1834) at this cell — a verified but not further explained fact (verified, Section 6).

The one-line-per-site fix. Change `c.cfct` from `0.1` to `0.2` at each of the three sites (Section 8), so both constants carry c_1 and the formula collapses to the documented single-constant separable form at every lag.

What remains open. Whether this defect changed any of paper 02's reported power, Type I error, or ranking conclusions has not been assessed. Every stored simulation output and every table or figure in `report.Rmd` and `supplement.Rmd` generated before this session's fix was computed under the halved cross-factor correlation. A rerun is a separate, larger task, tracked as Part 2 step 4 of `docs/60-trials-submission-gap-analysis.md`, and has not been performed as of this writing.

2. Scope

This note covers only the cross-factor correlation parameterization defect. Two other defects were found and fixed in the same driver in the same session (deterministic rather than randomized path allocation; silent rather than optional-stop non-positive-definite matrix repair), and are documented separately in `docs/60-trials-submission-gap-analysis.md` and in `CLAUDE.md`'s "Resolved 2026-10-21" entries, not here. This note also does not re-derive the refined process itself, which is paper 01's subject; it takes paper 01's Table `tab:spec` and `docs/series/ch10-refined- process.md` as the specification against which the code is checked.

The defect affects every call into `generate_data_multi_path()` from `simulate_cell()` or `simulate_cell_s7()`, which is to say every cell of paper 02's Tier 1 grid (`G1-G9` at the factorial grid of trial design, N, c_{bm}, $t_{1/2}$), every Tier 2 sensitivity block (S1-S11), the decay-shape sensitivity runs, and the higher-precision 24-cell paired rerun, from paper 02's first production run through 2026-10-21. It does not affect paper 01's own reported results, which do not use this driver (paper 01's own reproducibility pipeline is separate; see paper 01 Appendix B and its Reproducibility section).

3. The intended model

Paper 01's Table `tab:spec` (quoted):

> Cross-component correlation | separable, $c_1 \rho^{|w_t - w_s|}$; the Kronecker form applies to the correlation, since the PB SD scales with the expectancy | $c_1 = 0.2$

`docs/series/ch10-refined-process.md` states the same contrast against the published process (quoted, line 115):

> Cross-factor correlation ($t \neq s$) | constant $c_\times = 0.1$ | separable: $c_1 \rho^{|t-s|}$

and gives the separable form's defining property explicitly (line 215, quoted): "Separable AR(1), $\rho = 0.7$ per week, $c_1 = 0.2$ | choice: raises coupling ceiling from 0.256 to 0.480 (Hybrid)". The form is separable in the sense that the cross-factor correlation at any pair of visits, same or different, is one constant c_1 times the same decay $\rho^{|w_t - w_s|}$ that governs the within-factor AR(1) structure; at $w_t = w_s$ this is $c_1 \times 1 = c_1$ exactly, so a single parameter c_1 fully determines the cross-factor correlation at every timepoint pair.

In matrix terms (see `docs/63-why-correlation-fades-with-time-a-clinicians-guide.md`, Section 3, for the full derivation): the correlation structure over every (component, visit) pair is meant to be the Kronecker product $\Sigma = K_p \otimes K_t$, with K_t the AR(1) matrix ($[K_t]_{t,s} = \rho^{|w_t - w_s|}$) and K_p the 3×3 cross-component correlation matrix (TV, PB, BR), with 1's on the diagonal and the single constant c_1 off it. Entrywise, $\Sigma_{(c,t),(c',s)} = [K_p]_{c,c'} \cdot [K_t]_{t,s}$, which for $c \neq c'$ is exactly $c_1 \, \rho^{|w_t - w_s|}$. The code's two parameters, `c.cf1t` and `c.cfct`, are not two independent structural choices; they are two different code paths computing the *same* entry of K_p (the off-diagonal c_1) at two different ranges of lag ($\ell = 0$ and $\ell > 0$ respectively). Separability requires both code paths to return the same number. Section 4 shows they did not.

4. The code as it stood

`build_correlation_matrix()`, `implementations/tidyverse/R/functions.R`, lines 236-258 (inspected; unchanged by this fix, quoted here to show the two-parameter structure the caller must supply consistently):

```
for (factor2_idx in setdiff(seq_along(factors), factor_idx)) {
  other_factor <- factors[factor2_idx]

  n1_same <- paste(trial_design$timepoint_name,
                   current_factor, sep = ".")
  n2_same <- paste(trial_design$timepoint_name,
                   other_factor, sep = ".")
  idx1_same <- cbind(match(n1_same, labels), match(n2_same, labels))
  idx2_same <- cbind(idx1_same[, 2], idx1_same[, 1])
  correlations[idx1_same] <- model_param$c.cf1t
  correlations[idx2_same] <- model_param$c.cf1t

  if (num_timepoints > 1) {
    n1_lag <- paste(trial_design$timepoint_name[upper_pairs$p1],
                    current_factor, sep = ".")
    n2_lag <- paste(trial_design$timepoint_name[upper_pairs$p2],
                    other_factor, sep = ".")
    lag_vals <- model_param$c.cfct * ar_vals
    idx1_lag <- cbind(match(n1_lag, labels), match(n2_lag, labels))
    idx2_lag <- cbind(idx1_lag[, 2], idx1_lag[, 1])
    correlations[idx1_lag] <- lag_vals
    correlations[idx2_lag] <- lag_vals
  }
}
```

Here `ar_vals <- autocorrelation^time_gaps` with `autocorrelation` the within-factor ρ computed earlier in the same loop iteration (line 222), so `lag_vals` is exactly `c.cfct * rho^lag` as stated in Section 1. The package's `buildSigma()` (`R/generateData.R`, lines 322-344) implements the identical structure: `correlations[n1,n2] <- modelparam$c.cf1t` for same-timepoint pairs (lines 333-334) and `cfval <- modelparam$c.cfct * rho^tg` for different-timepoint pairs (line 342). Neither function was changed by this fix.

What was changed is the caller. `simulate_cell()` (`simulation-core.R`, before this session's fix, inspected from the version control history):

```
model_param <- list(
  N = cell$N,
  carryover_t1half = cell$t1half,
  carryover_form = cell$carryover_form,
  weibull_shape = cell$weibull_shape,
  dgp_architecture = cell$dgp_arch,
  c.tv = rho, c.pb = rho, c.br = rho,
  c.cf1t = 0.2, c.cfct = 0.1
)
```

`simulate_cell_s7()` and `26-co-dbc-fix-paired.R`'s `run_cell()` set the identical pair, `c.cf1t = 0.2, c.cfct = 0.1` (inspected).

5. Why the discrepancy is exactly a factor of 2

Write the intended cross-factor correlation at lag $\ell = |w_t - w_s|$ as $c_1\rho^\ell$, and the code's actual value as `c.cfct` $\times \rho^\ell$. The ratio is

$$\frac{\text{actual}}{\text{intended}} = \frac{\texttt{c.cfct} \times \rho^\ell}{c_1 \times \rho^\ell} = \frac{\texttt{c.cfct}}{c_1} = \frac{0.1}{0.2} = 0.5,$$

independent of ρ and of ℓ, for every $\ell > 0$ (derived). At $\ell = 0$ the same-timepoint branch of the code uses `c.cf1t` directly, which was set to $0.2 = c_1$ correctly, so same-visit cross-factor correlations were never affected. The defect is therefore not a diffuse numerical error but a clean, uniform halving of every cross-timepoint, cross-factor correlation entry, for as long as the driver has existed.

6. Worked example: Hybrid design, path 1

The Hybrid design's eight visits fall at weeks 4, 8, 9, 10, 11, 12, 16 and 20 (named `OL1, OL2, BD1, BD2, BD3, BD4, COd, COp` in the `tidyverse` implementation's trial design; verified).

Same timepoint (lag 0). TV-PB correlation at `OL1`: 0.2 under both the old and the corrected parameterization (verified) — unaffected, as expected.

Different timepoints (lag 4 weeks, `OL1` to `OL2`, weeks 4 to 8). At $\rho = 0.7$: intended $0.2 \times 0.7^4 = 0.04802$; code (before the fix) produced $0.1 \times 0.7^4 = 0.02401$; ratio 0.5 exactly (verified, matching Section 5's derivation).

General pattern across lags, $\rho = 0.7$ (derived from the verified ratio and the known decay):

Table 2: Intended versus actual cross-factor correlation by lag, $\rho = 0.7$; the lag-4 row is directly verified, the others derived from the same exact-factor-of-2 relationship

Lag (weeks)	ρ^{lag}	Intended ($c_1 = 0.2$)	Code (before fix, `c.cfct = 0.1`)
0	1.0000	0.2000	0.2000 (unaffected; uses `c.cf1t`)
1	0.7000	0.1400	0.0700
4	0.2401	0.0480	0.0240
8	0.0576	0.0115	0.0058
12	0.0138	0.0028	0.0014

Matrix-level effect. At the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ week), the Frobenius norm of the difference between the old and corrected correlation matrices is 0.625 (verified). Both the old and the corrected covariance matrices are positive definite at this cell; the minimum eigenvalue is 0.1025 under the old (weaker cross-correlation) parameterization and 0.1834 under the corrected one (verified). This is reported as an observed fact at one cell, not a general claim about how cross-factor correlation strength relates to positive-definiteness across the grid; no further mechanism is derived here.

7. Consequences

What changes. Every cross-timepoint, cross-factor correlation entry (TV-PB, TV-BR, PB-BR, at every pair of distinct visits) in every design the manuscript uses. In an eight-visit design this is 28 of the $\binom{8}{2} = 28$ possible cross-timepoint pairs per factor pair, all of them halved from their documented value; only the 8 same-timepoint pairs per factor pair were correct. With three factor pairs (TV-PB, TV-BR, PB-BR), the overwhelming majority of the model's cross-factor correlation structure was affected.

What does not change. The within-factor AR(1) correlations (`c.tv`, `c.pb`, `c.br`, all $\rho = 0.7$ in the affected cells), the biomarker-BR differential coupling (`c.bm`, the core interaction-carrying parameter under covariance moderation), graded coupling's off-drug decay, carryover on the drug-response mean, and the analysis model itself are all generated or specified by code paths independent of `c.cf1t`/`c.cfct` (inspected). The defect is confined to the background correlation structure among the nuisance/signal components with each other, not to the biomarker- treatment interaction the manuscript's power comparisons target directly.

Unknown downstream impact. Whether a halved cross-factor correlation among TV, PB and BR measurably changes any of paper 02's reported power, Type I error, or specification rankings is not established by this note. The covariance misspecification this introduces relative to the documented refined process could in principle affect the residual structure the `corCAR1` analysis model assumes, and therefore standard errors and power, in either direction; no simulation comparing old and corrected parameterizations head to head on the actual test statistics has been run as of this writing. This is the natural next step and is tracked separately (`docs/60-trials-submission-gap-analysis.md`, Part 2, step 4).

Scope of affected output. Every `.rds` output file and every table or figure in `analysis/report/02-carryover-sensitivit` and `supplement.Rmd` produced by a script that calls `simulate_cell()` or `simulate_cell_s7()` was generated under the halved cross-factor correlation, from paper 02's first production run through 2026-10-21.

8. The fix

Three sites, each changing `c.cfct` from `0.1` to `0.2`:

`simulation-core.R, simulate_cell()` (model_param list, previously ending `c.cf1t = 0.2, c.cfct = 0.1`):

```
c.tv = rho, c.pb = rho, c.br = rho,
c.cf1t = 0.2, c.cfct = 0.2
```

simulation-core.R, simulate_cell_s7() (same change, same pattern).

26-co-dbc-fix-paired.R, run_cell() (same change):

```
c.tv = 0.7, c.pb = 0.7, c.br = 0.7, c.cf1t = 0.2, c.cfct = 0.2
```

No change was made to `build_correlation_matrix()` or `buildSigma()`; both already correctly implement the general two-constant structure and require only that the caller supply matching constants to recover the documented single-c_1 form.

Verification that the fix restores the documented ratio of 1 (not 0.5) at lag > 0: see Section 9's reproduction commands, which confirm the lag-4, $\rho = 0.7$ cell now equals $0.2 \times 0.7^4 = 0.04802$ exactly, and that the lag-0 value is unchanged at 0.2 under both the old and new parameterizations.

Several older one-off diagnostic scripts in the same directory (`diagnose-null-type1*.R`, `check-null-architecture-equival` `06-cr2-recalibration.R`, `diagnose-alt-cell-cr2.R`) and the earlier `quick-sim-prototype*.R` exploratory scripts still construct `model_param` directly with the old, or in some cases a different (`c.cf1t = 0.1, c.cfct = 0.05`), pair of constants. None of these were found to feed the manuscript's reported tables or figures directly, and they were left unchanged in this pass; this was not exhaustively verified for every script in the directory, so a later sweep should confirm none of their output is cited before relying on that claim.

9. How to verify independently

Run in a checkout of this repository, from the repository root:

```
library(corpcor)
source("implementations/tidyverse/R/functions.R")
source("analysis/scripts/carryover-sensitivity/simulation-core.R")

design_set <- build_design_set("Hybrid")
td <- design_set[[1]]
rp <- default_resp_param(); bp <- default_baseline_param()

mp_new <- list(N = 70, carryover_t1half = 1.0,
               carryover_form = "exponential", weibull_shape = 1,
               dgp_architecture = "mvn", c.bm = 0.45,
               c.tv = 0.7, c.pb = 0.7, c.br = 0.7,
               c.cf1t = 0.2, c.cfct = 0.2)       # corrected
mp_old <- mp_new; mp_old$c.cfct <- 0.1           # as the driver stood before this fix

s_new <- build_sigma_matrix(mp_new, rp, bp, td, dgp_architecture = "mvn")
s_old <- build_sigma_matrix(mp_old, rp, bp, td, dgp_architecture = "mvn")

tdat <- prepare_trial_data(td)
nm <- td$timepoint_name

# lag 0: unaffected
i1 <- paste0(nm[1], ".tv"); i2 <- paste0(nm[1], ".pb")
s_old$correlations[i1, i2]; s_new$correlations[i1, i2]   # both 0.2
```

```r
# lag 4 weeks (OL1 to OL2): halved before the fix
j1 <- paste0(nm[1], ".tv"); j2 <- paste0(nm[2], ".pb")
s_old$correlations[j1, j2]   # 0.02401
s_new$correlations[j1, j2]   # 0.04802

norm(s_old$correlations - s_new$correlations, type = "F")  # 0.625
is.positive.definite(s_old$sigma); is.positive.definite(s_new$sigma)  # both TRUE
min(eigen(s_old$sigma, only.values = TRUE)$values)   # 0.1025
min(eigen(s_new$sigma, only.values = TRUE)$values)   # 0.1834
```

All values above reproduce exactly (verified, this session).

10. Limitations

- **No downstream power or Type I error impact has been assessed.** This note documents the DGP-level defect and its exact size, not its effect on any of paper 02's specific reported results. See Section 7.
- **Positive-definiteness and eigenvalue behavior were checked at one reference cell only** (Hybrid, $c_{bm} = 0.45$, $t_{1/2} = 1.0$), not swept across the full grid of designs, sample sizes, effect sizes and half-lives paper 02 uses.
- **Older diagnostic and prototype scripts were not exhaustively audited**; Section 8 names the ones found still using stale constants and states only that they were not found to feed the manuscript directly, not that this was verified for every script.
- **This note does not address the driver's other two defects** (deterministic path allocation; silent non-PD repair), which were found and fixed in the same session but are out of scope here; see `docs/60-trials-submission-gap-analysis.md` and `CLAUDE.md`.

11. Reproducibility

Section 9 gives the exact commands used to produce every verified number in this note. No stored data files are required; the comparison is deterministic given the model parameters shown (no random seed is involved, since `build_sigma_matrix()` only constructs the covariance matrix, not simulated draws from it).

12. References

- Paper 01, Table `tab:spec` (`analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`).
- `docs/series/ch10-refined-process.md`, the refined process's canonical specification.
- `docs/60-trials-submission-gap-analysis.md`, Part 2, which first identified this defect (step 1) as part of a broader session auditing paper 02 against the refined process.
- `CLAUDE.md` (pmsimstats-ng project file), "Resolved 2026-10-21" entries, for this defect's sibling fixes (path allocation, non-PD handling) in the same driver.
- `docs/53-carryover-recursion-in-published-code.md` and `docs/55-step-coupling-in-published-code.md`, whose structure this note follows, though both describe defects in Hendrickson et al.'s published code rather than in this project's own.